Project Planning

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Project Title

Personalized Travel Itinerary Planner

Project Timeline & Milestones

Phase 1: Foundation (Week 1)

Task	Duration	Status
Set up Supabase project and database schema	1 day	Completed
Design and implement RLS policies and helper functions	1 day	Completed
Set up FastAPI backend project structure	1 day	Completed
Implement JWT authentication with JWKS verification	1 day	Completed
Create core configuration and error handling	1 day	Completed

Phase 2: Backend API Development (Week 2)

Task	Duration	Status
Implement Trip CRUD endpoints (create, list, get, update, delete)	2 days	Completed
Implement Activity CRUD endpoints (update, delete)	1 day	Completed
Build AI service with Gemini API integration	2 days	Completed
Add Groq API fallback mechanism	0.5 days	Completed
Implement prompt engineering for itinerary generation	1 day	Completed
Build generation service (trip + day orchestration)	1 day	Completed

Phase 3: Frontend Development (Week 3)

Task	Duration	Status
Set up React + Vite + TypeScript project	0.5 days	Completed
Implement Login and Signup pages with Supabase Auth	1 day	Completed
Build Dashboard page with trip listing	1 day	Completed
Create New Trip form with Zod validation	1 day	Completed
Develop Trip Detail page with day-by-day view	2 days	Completed
Add Activity Edit Dialog	1 day	Completed

Phase 4: Advanced Features (Week 4)

Task	Duration	Status
Implement drag-and-drop activity reordering (dnd-kit)	1 day	Completed
Add day regeneration feature	1 day	Completed
Implement activity reorder API with time slot swapping	1 day	Completed
Add loading states and error handling across UI	1 day	Completed
Write unit tests (schemas and trip service)	1 day	Completed

Phase 5: Deployment & Documentation (Week 5)

Task	Duration	Status
Deploy frontend to Vercel	0.5 days	Completed
Deploy backend to Render	0.5 days	Completed
Configure environment variables and CORS	0.5 days	Completed
End-to-end testing with deployed services	1 day	Completed
Prepare project documentation	2 days	Completed

Resource Allocation

Resource	Role	Contribution
Developer	Full-stack development	Backend (FastAPI), Frontend (React), Database (Supabase), AI integration
Supabase	BaaS Platform	PostgreSQL database, Authentication, Row-Level Security

Resource	Role	Contribution
Gemini API	AI Service	Primary LLM for itinerary generation
Groq API	AI Service (Fallback)	Secondary LLM when Gemini is unavailable
Vercel	Hosting	Frontend deployment with CDN
Render	Hosting	Backend deployment with managed containers

Risk Assessment

Risk	Probability	Impact	Mitigation
LLM returns invalid JSON	Medium	High	Pydantic validation + retry once + Groq fallback
LLM rate limiting (429)	Medium	Medium	Retry with 2s delay (up to 3 times) + Groq fallback
Supabase downtime	Low	High	Supabase provides 99.9% SLA; monitoring via health endpoint
Security breach	Low	Critical	RLS policies, JWT verification, CORS restrictions, input validation
Budget overestimation by AI	Medium	Low	AI prompt explicitly sets budget limits; user can edit costs